Lecture

Introduction to Logic (CS), Lecture 5

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Today's Lecture

1. Overview and Introduction
2. Full First-Order Logic
3. Translating with Quantifiers
4. Counting with quantifiers

This Week

Language of first-order logic (FOL)

- Atomic sentences ✓
- Boolean connectives ✓
- Conditionals ✓
- **Quantifiers (today):** new language constructs, translation.

Learning goals: At the end of this week, you are able to

- construct well-formed sentences with quantifiers
- translate quantified English sentences into FOL

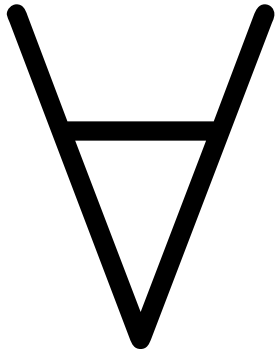
Quantifiers in English

- All
- Many
- Some
- No
- Few

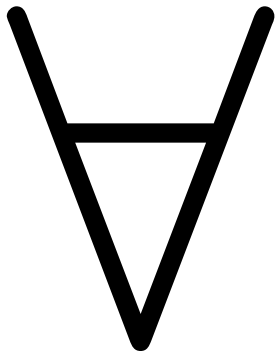
Quantifiers in English

- All courses are fun.
- Many students work hard.
- Some exams are doable.
- No teacher is perfect.
- Few books are dull.

Universal quantifier (“for all ...”)

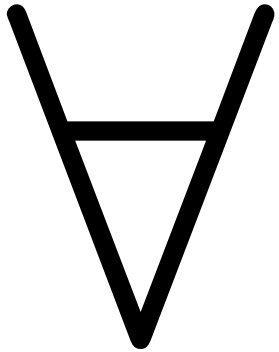


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Example: $\forall x\ \mathrm{Happy}(x)$

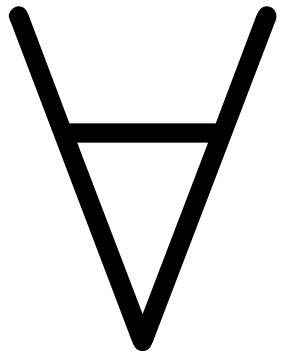
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- Natural reading: *Everyone is happy*
- Who is “everyone”? Or is it “everything”?

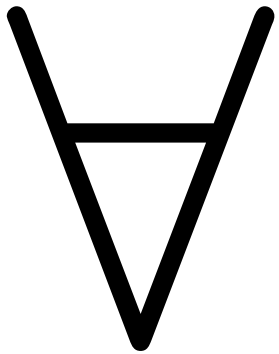
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Example: $\forall x\ \mathrm{Happy}(x)$

- x is universally quantified
- Natural reading: *Everyone is happy*
- Who is “everyone”? Or is it “everything”?
- Precise reading: *All objects in the domain of discourse have the property of being happy.*

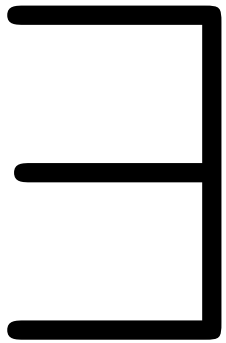
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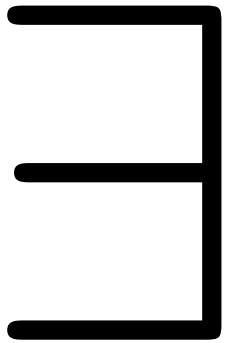
Use parenthesis to disambiguate the scope:

$$\forall x\,(\mathit{Happy}(x) \wedge \mathit{Satisfied}(x))$$

Existential quantifier (“there exists ...”)

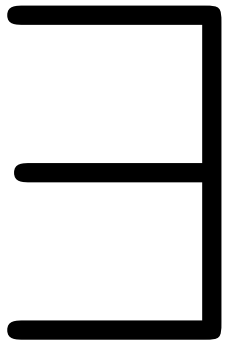


Existential quantifier (“there exists ...”)



Example: $\exists x\ \mathrm{Tall}(x)$

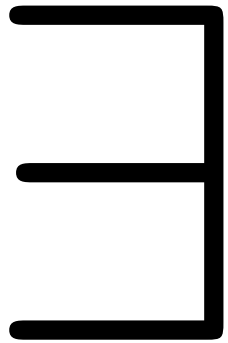
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Example: $\exists x\ \mathrm{Tall}(x)$

- Natural reading: *Someone is tall*

Existential quantifier ("there exists ...")



Example: $\exists x\ \mathrm{Tall}(x)$

- Natural reading: *Someone is tall*
- Precise reading: *Some object in the domain of discourse has the property of being tall.*
- x is existentially quantified

First application of quantifiers: Syllogisms

From Aristotle (384-322 BC) till the 19th century, logic was about *syllogisms*.

All rabbits have fur
Some animals are rabbits
Therefore, some animals have fur

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Based on 4 types of categorical propositions:

- universal affirmative: Every S is a P
- particular affirmative: Some S is a P
- universal negative: No S is a P
- particular negative: Some S is not a P

First application of quantifiers: Syllogisms

Categorical propositions	are expressible with quantifiers
universal affirmative:	Every S is a P
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can be rewritten as

"For every thing, if that thing is S, then it is also P"

In particular, $\forall x(S(x) \wedge P(x))$ is incorrect!.

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Consider the scenario in which there are two animals: Bob and Jane. Bob is a dinosaur and a bird, and Jane is a cat. Then we can safely say that in that situation *every dinosuar is a bird*:

$$\forall x(Dino(x) \rightarrow Bird(x))$$

holds, but the incorrect version does not:

$$\forall x(Dino(x) \wedge Bird(x))$$

Some Translations into FOL

Assume the domain of discourse contain all courses, exams, teachers, students, ...

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$$\exists y\,(\text{Exam}(y) \wedge \text{Doable}(y))$$
- No teacher is perfect.
  - $\forall x\,(\text{Teacher}(x) \rightarrow \neg\,\text{Perfect}(x))$, or
  - $\neg\exists x\,(\text{Teacher}(x) \wedge \text{Perfect}(x))$

NB: “Many” and “Few” are not expressible in FOL.

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FOL

Ingredients of the language of FOL up to now:

- terms (to denote objects) built from
  - constants a, b, c, ...
  - functions f, g, h, +, ...
  - *E.g.*: a, f(a), f(f(a)), g(b,c), g(g(f(c),c),f(g(a,b))), ...
- sentences (that have a truth value) built from
  - predicates (applied to terms) P, Q, A, B, ...
  - connectives $\bot, \neg, \wedge, \vee, \rightarrow, \leftrightarrow$
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We now add:

- quantifiers $\forall, \exists$
- variables x, y, z, ...
- well-formed formulas (wffs) P(x), A(f(y),a) $\vee$ B(x), ...

Variables and Atomic Well-Formed Formulas

- We assume an unlimited supply of **variables**:
  $x, y, z, u, \ldots, x_0, x_1, x_2, x_3, \ldots, y_0, y_1, y_2, y_3, \ldots$.
- Notational convention: letters from the end of the alphabet. (We reserve letters from the beginning of the alphabet for constants.)
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  - father(x), mother(father(x)) etc. where x is a variable.
  - Cube(y), Larger(y,c) etc. where y is a variable and c is a constant.

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- An **atomic well-formed formula (atomic wff)** is built using only variables, constants, function symbols and predicate symbols (no connectives/conditionals/quantifiers), using the correct arities

Well-Formed Formulas (wffs)

We define the collection of **well-formed formulas (wffs)** as follows:

- If P is an atomic wff, then it is a wff.
- If P is a wff, so is $\neg P$.
- If $P_1, \ldots, P_n$ are wffs, so is $(P_1 \wedge \ldots \wedge P_n)$.
- If $P_1, \ldots, P_n$ are wffs, so is $(P_1 \vee \ldots \vee P_n)$.
- If P and Q are wffs, so is $(P \rightarrow Q)$ and $(P \leftrightarrow Q)$.
- If P is a wff and x is a variable then $\forall x\, P$ and $\exists x\, P$ are wffs.
- Nothing else is a wff.

We usually omit outermost parentheses.

Examples of Well-Formed Formulas

Examples of wffs:

- $\exists x\,(\mathit{Small}(x) \wedge \mathit{Blue}(x) \wedge \mathit{Cube}(x))$
- $\forall x\,(\mathit{Cube}(x) \rightarrow \mathit{Small}(x))$
- $\exists x\,\mathit{Cube}(x) \wedge \exists x\,\mathit{Small}(x)$

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In the last example, x occurs as a free variable.

Free and Bound Variable Occurrences

- In $\forall x\, P$, the quantifer $\forall x$ *binds* all free occurrences of x in P.
  - Example: In $\forall x\,(R(x, y) \wedge S(x))$, $\forall x$ binds x in $R(x, y) \wedge S(x)$.
  - Think of variables and scopes in programming (`int f(int x) {...}`)

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- Similarly for $\exists x$.
- (See LPL Ch 9.3 for a more precise definition.)

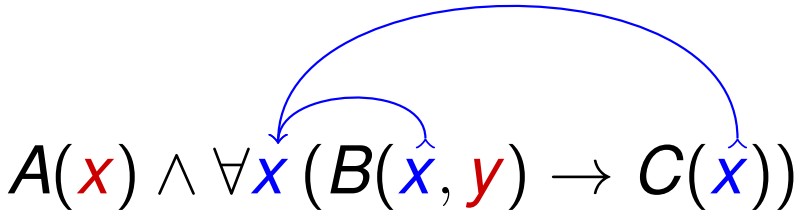
Free and Bound Variable Occurrences

- Example:

$$A(x) \wedge \forall x\,(B(x, y) \rightarrow C(x))$$

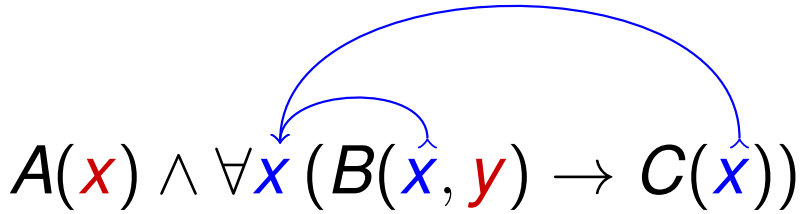
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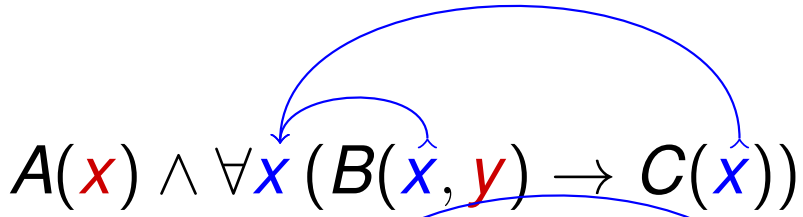


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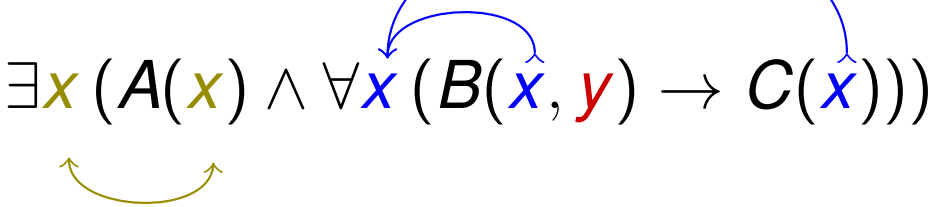
$$\exists x\,(A(x) \wedge \forall x\,(B(x, y) \rightarrow C(x)))$$

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Note: a variable can occur both free and bound in the same formula.

Free and Bound Variable Occurrences

The notion of variable scope is very important in programming:

```
>>> def f(x):
...     def g(x):
...         return x-1
...     return (g(x+3))
...
>>> f(3)
5
>>> x = 2
>>> f(x)
4
```

Well-formed sentences

Some formulas we can read off readily:

$$\forall x(\mathit{Cube}(x) \rightarrow \exists y(\mathit{NextTo}(x, y)))$$

other are not possible to read off as an English sentence:

$$\mathit{Cube}(x) \wedge \mathit{Tall}(x)$$

What is the main issue?

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Definition. A (well-formed) sentence is a wff in which all variable occurrences are bound by a quantifier.

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Exercise: Which of the following are sentences? (x, y are variables, a, b are constants):

1. $\exists x\, \mathit{Blue}(x) \wedge \mathit{Cube}(x)$
2. $\forall x\, \exists y\, \mathit{Smaller}(y, x) \wedge \forall x\, \mathit{LeftOf}(x, y)$
3. $\exists a\, \mathit{Blue}(a) \wedge \mathit{Larger}(a, b)$
4. $\forall x\, \mathit{Cube}(x) \rightarrow x = a$
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2. $\forall x\, \exists y\, Smaller(y, x) \wedge \forall x\, LeftOf(x, y)$
   $\forall x\, \exists y\,(Smaller(y, x) \wedge \forall x\, LeftOf(x, y))$
3. $\exists a\, Blue(a) \wedge Larger(a, b)$ $\quad \exists x\,(Blue(x) \wedge Larger(x, b))$
4. $\forall x\, Cube(x) \rightarrow x = a$ $\quad \forall x\,(Cube(x) \rightarrow x = a)$
5. $\forall x\,(a = a)$

Well-formed sentences

Definition. A (well-formed) sentence is a wff in which all variable occurrences are bound by a quantifier.

Exercise: Which of the following are sentences? (x, y are variables, a, b are constants):

1. $\exists x\, Blue(x) \wedge Cube(x)$ $\quad \exists x\,(Blue(x) \wedge Cube(x))$
2. $\forall x\, \exists y\, Smaller(y, x) \wedge \forall x\, LeftOf(x, y)$
   $\forall x\, \exists y\,(Smaller(y, x) \wedge \forall x\, LeftOf(x, y))$
3. $\exists a\, Blue(a) \wedge Larger(a, b)$ $\quad \exists x\,(Blue(x) \wedge Larger(x, b))$
4. $\forall x\, Cube(x) \rightarrow x = a$ $\quad \forall x\,(Cube(x) \rightarrow x = a)$
5. $\forall x\,(a = a)$ ✓

Semantics of $\forall$, $\exists$ (satisfaction by objects)

Let $P(x)$ be a wff with x as the only variable occurring freely (possibly zero times). Examples:

- $Cube(x) \wedge Small(x)$.
- $\exists y\, Cube(y) \wedge Small(x)$.
- $a = a$.

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- $\exists y\, Cube(y) \wedge Small(x)$.
- $a = a$.

Definition. We say that an object d from the domain of discourse satisfies $P(x)$ if the sentence $P(n)$ is true where n is a *fresh and temporary name for* d.

Semantics of ∀, ∃

The domain of discourse is the collection of objects that ∀ and ∃ quantify over. We assume the domain of discourse is nonempty.

Semantics of $\forall$, $\exists$

The domain of discourse is the collection of objects that $\forall$ and $\exists$ quantify over. We assume the domain of discourse is nonempty.

Definition

- A sentence $\forall x\, P(x)$ is true if and only if

  every object in the domain of discourse satisfies $P(x)$.

Semantics of $\forall$, $\exists$

The domain of discourse is the collection of objects that $\forall$ and $\exists$ quantify over. We assume the domain of discourse is nonempty.

Definition

- A sentence $\forall x\, P(x)$ is true if and only if

  every object in the domain of discourse satisfies $P(x)$.
- A sentence $\exists x\, P(x)$ is true if and only if

  at least one object in the domain of discourse satisfies $P(x)$.

Quantifiers $\forall$, $\exists$ as generalisations of $\wedge$, $\vee$

Suppose every object in the domain of discourse has a name:

$$a, b, c, \ldots$$

Then

$\forall x\ P(x)$ *corresponds to* $P(a) \wedge P(b) \wedge P(c) \wedge \ldots$

$\exists x\ P(x)$ *corresponds to* $P(a) \vee P(b) \vee P(c) \vee \ldots$

(This intuition will be useful for the proof rules.)

Today's Lecture

1 Overview and Introduction

2 Full First-Order Logic

3 Translating with Quantifiers

4 Counting with quantifiers

Translating step-by-step

Someone feels a connection to no one.

Translating step-by-step

Someone feels a connection to no one.

$\exists x$ *x* feels a connection to no one.

Translating step-by-step

Someone feels a connection to no one.

$\exists x$ x feels a connection to no one.

$\exists x \neg$ (x feels a connection to someone).

Translating step-by-step

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$\exists x \neg \exists y$ (x feels a connection to y).

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$\exists x$ x feels a connection to no one.

$\exists x \neg$ (x feels a connection to someone).

$\exists x \neg \exists y$ (x feels a connection to y).

$\exists x \neg \exists y\ C(x, y)$

Translation key:
$C(x, y)$: x feels a connection to y.

Translating step-by-step (tricky)

Between every two cubes there is a tetrahedron.

Translating step-by-step (tricky)

Between every two cubes there is a tetrahedron.

Reformulate:
For every two cubes, there is a tetrahedron between them.

Translating step-by-step (tricky)

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Reformulate:
For every two cubes, there is a tetrahedron between them.

$\forall x \forall y((Cube(x) \land Cube(y)) \rightarrow \exists z(Tet(z) \land Between(z, x, y)))$

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$\forall x \forall y((Cube(x) \land Cube(y) \land (x \neq y)) \rightarrow$
$\exists z(Tet(z) \land Between(z, x, y)))$

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If anyone/someone knows the answer, (s)he may say so.

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$$\exists x K(x) \rightarrow S(x)?$$

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$$\exists x K(x) \rightarrow S(x)?$$

Problem: the x in $S(x)$ is not bound by $\exists x$.
Remedy: reformulate the sentence to be translated.

Translating step-by-step (tricky)

If anyone/someone knows the answer, (s)he may say so.

$$\exists x K(x) \rightarrow S(x)?$$

Problem: the x in $S(x)$ is not bound by $\exists x$.
Remedy: reformulate the sentence to be translated.

Everyone who knows the answer may say so.

$$\forall x(K(x) \rightarrow S(x))$$

where
$K(x)$: x knows the answer
$S(x)$: x may say that they know the answer.

Translating step-by-step (tricky bits)

Why does $\exists x(K(x) \rightarrow S(x))$ not work?

Translating step-by-step (tricky bits)

Why does $\exists x(K(x) \to S(x))$ not work?

Suppose domain of discourse consists of Ann and Bob, and:

- Ann knows the answer ($K(a)$), and she may say so ($S(a)$).
- Bob knows the answer ($K(b)$), but he may not say so ($\neg S(b)$).

Then Ann satisfies $K(x) \to S(x)$, but Bob does not.
Hence $\exists x(K(x) \to S(x))$ is true, but $\forall x(K(x) \to S(x))$ (correctly) is false.

“Donkey” sentences

Everyone who owns a dog, must take good care of it.

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$\forall x$ (x owns a dog $\rightarrow$ x must take good care of it.)

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$\forall x(\exists y($ y is a dog $\wedge$ x owns $y) \rightarrow$ x must take good care of y.)

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Problem: the final y is not bound by $\exists y$
Reformulation:

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Everyone must take good care of all dogs (s)he owns.

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Problem: the final y is not bound by $\exists y$
Reformulation:

Everyone must take good care of all dogs (s)he owns.

$\forall x \forall y((Dog(y) \wedge Owns(x, y)) \rightarrow MustTakeGoodCare(x, y))$

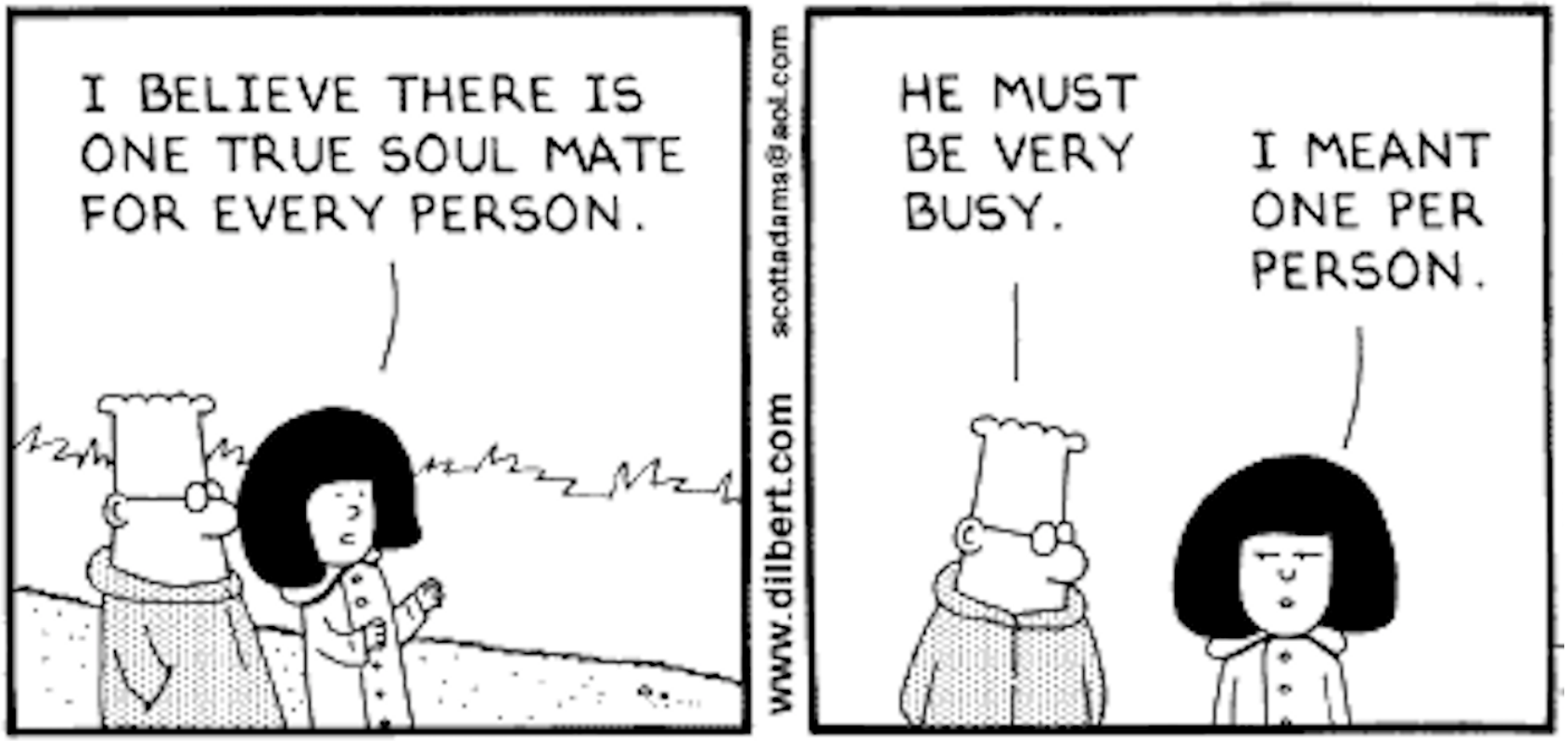
Translating step-by-step (tricky bits)

Scoping (and quantifiers) are complicated and can often be ambiguous in natural languages.

Thus, it is quite a bit harder to translate then just in PL.

But that makes it only more valuable.

Mixing ∀ and ∃


I BELIEVE THERE IS ONE TRUE SOUL MATE FOR EVERY PERSON.
HE MUST BE VERY BUSY.
I MEANT ONE PER PERSON.
www.dilbert.com scottadams@aol.com

Mixing $\forall$ and $\exists$

Meaning of $\forall x \exists y A(x, y)$:

Mixing $\forall$ and $\exists$

Meaning of $\forall x \exists y A(x, y)$:

$A(x, y)$: x loves y.

Mixing $\forall$ and $\exists$

Meaning of $\forall x \exists y A(x, y)$:

$A(x, y)$: x loves y. "Everyone loves someone"

Mixing $\forall$ and $\exists$

Meaning of $\forall x \exists y A(x, y)$:

$A(x, y)$: x loves y. "Everyone loves someone"

$A(x, y)$: x is the soulmate of y.

Mixing $\forall$ and $\exists$

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Observe: y may depend on x; different x's may require different y's to satisfy A.

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Now a *single* y must satisfy $A(x, y)$ for all x.

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Now a *single* y must satisfy $A(x, y)$ for all x.

When mixing $\forall$ and $\exists$, order matters.

Mixing $\forall$ and $\exists$

We have:

- $\exists y \forall x A(x, y) \Rightarrow \forall x \exists y A(x, y)$, but
- $\forall x \exists y A(x, y) \not\Rightarrow \exists y \forall x A(x, y)$

Mixing $\forall$ and $\exists$

We have:

- $\exists y \forall x A(x, y) \Rightarrow \forall x \exists y A(x, y)$, but
- $\forall x \exists y A(x, y) \not\Rightarrow \exists y \forall x A(x, y)$

(We will make logical consequence precise in week 7.)

Example: Take $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ as the domain of discourse, then:

$$\forall x \exists y\ (x < y), \quad \text{but } \textbf{not} \quad \exists y \forall x\ (x < y)$$

Today's Lecture

1 Overview and Introduction

2 Full First-Order Logic

3 Translating with Quantifiers

4 Counting with quantifiers

Unexpressable and expressable quantifiers

Not all natural language quantifiers are expressible in FOL: “many”, “few”, etc...
(We cannot prove this formally yet)

But some others can be expressed in terms of $\forall$ and $\exists$: “exactly one...”, “at most two...”, ec

Expressing “at least 1”, “at most 1”, “exactly 1”

- *There is at least one* x *with* $A(x)$: $\exists x A(x)$

Expressing “at least 1”, “at most 1”, “exactly 1”

- *There is at least one* x *with* $A(x)$: $\exists x A(x)$
- *There is at most one* x *with* $A(x)$:

Expressing “at least 1”, “at most 1”, “exactly 1”

- *There is at least one x with $A(x)$:* $\exists x A(x)$
- *There is at most one x with $A(x)$*: Reformulate: whenever two elements satisfy A, they have to be equal. In other words, *whenever $A(x)$ and $A(y)$, we have $x = y$*

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- *There is at least one* x *with* $A(x)$: $\exists x A(x)$
- *There is at most one* x *with* $A(x)$: Reformulate: whenever two elements satisfy A, they have to be equal. In other words, *whenever* $A(x)$ *and* $A(y)$*, we have* $x = y$
  So we get: $\forall x \forall y((A(x) \wedge A(y)) \rightarrow x = y)$.

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Expressing “at least 1”, “at most 1”, “exactly 1”

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- *There is at most one x with $A(x)$*: Reformulate: whenever two elements satisfy A, they have to be equal. In other words, *whenever $A(x)$ and $A(y)$, we have $x = y$*
  So we get: $\forall x \forall y((A(x) \wedge A(y)) \rightarrow x = y)$.
- *There is exactly one x with $A(x)$:*

$$\exists! x A(x) := \exists x A(x) \wedge \forall x \forall y((A(x) \wedge A(y)) \rightarrow x = y)$$

or, equivalently, $\exists x \forall y(A(y) \leftrightarrow x = y)$

Expressing “at least 2”, “at most 2”, “exactly 2”

- *There are at least two x with $A(x)$: ?*

Expressing “at least 2”, “at most 2”, “exactly 2”

- *There are at least two x with $A(x)$*: ?
  Use two variables x, y and add the condition $x \neq y$:

$$\exists x \exists y (x \neq y \wedge A(x) \wedge A(y))$$

Expressing “at least 2”, “at most 2”, “exactly 2”

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  Use two variables x, y and add the condition $x \neq y$:

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- *There are at most two x with $A(x)$*: ?

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$$\exists x \exists y (x \neq y \wedge A(x) \wedge A(y))$$

- *There are at most two x with $A(x)$*: ? Reformulate: whenever three objects satisfy A, two of them are equal:

$$\forall x \forall y \forall z ((A(x) \wedge A(y) \wedge A(z)) \rightarrow (x = y \vee x = z \vee y = z))$$

Expressing “at least 2”, “at most 2”, “exactly 2”

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- *There are exactly two x with $A(x)$*: ?

Expressing "at least 2", "at most 2", "exactly 2"

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- *There are exactly two x with $A(x)$*: ?

$$\exists x \exists y (x \neq y \wedge A(x) \wedge A(y) \wedge \forall z (A(z) \rightarrow (x = z \vee y = z)))$$

or

$$\exists x \exists y (x \neq y \wedge \forall z (A(z) \leftrightarrow (x = z \vee y = z)))$$

Expressing “at least *n*”, etc

Obviously this generalizes to “at least *n*...”, and “exactly *n*...” for each concrete $n \in \mathbb{N}$.

But this *n* has to be known in advance! This suggest why we cannot use the same method for expressing “at least half of the people...” or “majority of objects...”

Wrapping up

Learning goals: At the end of this week, you are able to

- construct well-formed sentences with quantifiers
- express quantified English sentences in FOL

Week 5 material

- Reading, LPL Chapter 9
- Exercises, see Brightspace.

Next up:

- Formal proof rules for quantifiers (the logic of quantifiers).
- First-order models (formal semantics for FOL)